Chapter 8
Artificial Intelligence: A Smart and Empowering Approach to Women's Safety

Varkha K. Jewani
K.C. College, Mumbai, India

Prafulla E. Ajmire
G.S. College, Khamgaon, India

Suhashini Chaurasia
https://orcid.org/0000-0002-7443-0105
S.S. Maniar College of Computer and Management, Nagpur, India

Geeta N. Brijwani
K.C. College, Thane, India

ABSTRACT

In today's society, women's safety and empowerment are top priorities. Artificial intelligence (AI) integration offers a revolutionary means of resolving these problems. This abstract examines a clever and empowering strategy that makes use of AI technologies to improve the safety of women.AI-powered personal safety applications dramatically improve individual security by providing real-time location monitoring, emergency notifications, and connectivity with trusted contacts. The use of AI algorithms in predictive policing detects high-risk regions and patterns of violence against women, allowing for tailored law enforcement responses. AI-enabled safety chatbots and hotlines offer a secure environment for reporting occurrences and provide details on one's legal rights and available assistance options. Platforms for reporting and crowdsourcing data enable women to contribute to data-driven safety efforts, enabling more efficient responses. Initiatives for community interaction powered by AI raise awareness and enable quick solutions to safety issues.

DOI: 10.4018/979-8-3693-2679-4.ch008

1. INTRODUCTION

Women's safety continues to be a pressing global concern in a time of unheard-of technical developments. It is the obligation of society to ensure women's safety and well-being as they traverse a variety of locations and situations, which calls for creative and practical solutions. Artificial intelligence's branch of machine learning is showing promise as a powerful tool for enhancing women's safety. Women's safety involves a variety of issues, such as protecting their physical safety, being shielded from harassment, preventing domestic violence, and providing safe travel experiences. Due to their pervasiveness, solutions must be all-encompassing and flexible—able to react quickly to new threats and offer support when it is required (Choudhary, 2017). In addition to ensuring the security and empowerment of women a fundamental human right, but it is also a cornerstone of creating societies that are fair, inclusive, and progressive. However, women are still frequently the targets of harassment, abuse, and discrimination, which seriously jeopardizes their general well-being and personal safety (Kaur et al., 2016). There is a fantastic chance to use artificial intelligence (AI) to develop a clever and powerful strategy to solve women's safety issues in the current period as technology continues to transform our society. With its capacity to absorb enormous volumes of data, identify trends, and make choices in real-time, AI has the potential to completely change how we approach the complex problem of women's safety. This introduction explores the idea of using artificial intelligence (AI) as a change agent, emphasizing how it may empower women by boosting their safety, giving them access to priceless resources, and promoting a sense of security in their daily lives (IEEE SCOPES International Conference et al., 2016). While numerous social, legal, and community-based programs have traditionally been used to address women's safety, AI adds a fresh perspective to this effort. We can develop cutting-edge solutions that are not only responsive but also proactive in avoiding and addressing risks to the safety of women by utilizing AI technologies. There are numerous uses for these AI-powered products, including community engagement platforms, safety chatbots, and personal safety apps to predictive policing, and beyond.

This clever and inspiring strategy acknowledges that protecting women is a shared responsibility that necessitates the incorporation of cutting-edge technology and community involvement. It is not just the job of law enforcement or community organizations. It considers the particular requirements and experiences of women in various geographic, cultural, and socioeconomic situations to make sure AI-driven solutions are usable, sensitive to cultural differences, and suited to the various obstacles that women may encounter. In this investigation of "A Smart and Empowering Approach for Women's Safety Using Artificial Intelligence," we will go further into the numerous components of this method. We will go through the essential elements, uses, and advantages of artificial intelligence (AI) in boosting women's safety, as well as important issues including data privacy, community involvement, and the importance of education and awareness. We set out on a path to create a safer, more inclusive society where women can thrive with confidence and empowerment by embracing this junction of technology and social development.

2. LITERATURE REVIEW

An increasing body of research and activities are aimed at utilizing AI technologies to solve women's safety problems, according to a literature review on "A Smart and Empowering Approach for Women's

Safety Using Artificial Intelligence". The following survey highlights significant discoveries and developments in the subject and covers studies, projects, and publications up to my most recent knowledge:

A mechanism was proposed by Charranzhou—fly utilizing GPS-enabled mobile devices. The author used PR technologies and data-oriented machine code to model a computer and determine the speed, distance, and traveling directions (Chougula et al., 2014; Sangoi, 2014). The identity of the device owner is defined and categorized using these characteristics. The author verified PR technologies in the random forest and consistently tracked the range from moving targets. To achieve their goals, PravinKshirsagar et al. (Chougula et al., 2014) have defined a variety of neural networks, including the activation functions of one layer, layered perceptions, RBF, NEP (Pnn), Grnn, and others. A small number of simple runtime data and structures were shown to benefit from this conventional neural network (Chand et al., 2015; Pravin & Akojwar, 2016d). To ensure that women never feel exposed to problems or challenges in society, A.H. Ansari proposed new protective technologies for women using GSM & GPS. With a raspberry pi, GSM, GPS, and motion detector, women's security is guaranteed. The only button on the device needs to be pushed whenever a woman feels threatened (Pradeep et al., 2015; Sangoi, 2014). In that case, the GPS would map the location of the women and send a message of emergency to the contacts and control room that were saved by GSM.GSM and GPS models have been suggested by B. Vijayalakshmi as a way to improve female protection. An extremely small device with a buzzer and a microcontroller is created and may be installed on the strip or the monitor. If necessary, the woman can utilize this device to send SMS warnings to 5 members by pressing the buzzer. However, it is unable to send out an SMS alert right away. People contact more slowly while in a crisis condition (Deshpande & Kalita, 2014). A technique that combines photos with metadata to categorize people's locations was identified by Rameshkumar.P. A GPS mapper is used to pinpoint a person's location using background data like images and videos. An individual who has uploaded photos to social media can identify elevation, latitude, and position using a GPS mapper. But it can't make an image of someone who isn't sharing it on social media (Chand et al., 2015).Since the market for mobile devices has grown quickly, gadgets can now be utilized for a variety of different types of protection in the modern environment (Pravin & Akojwar, 2016d). The guidelines for women's workplace safety are primarily categorized under the following four headings: mechanical, ecological, administrative, and professional. Each head offers worthy and aspirational proposals. This emphasizes the defensive capacity of female employees in an organization (Pradeep et al., 2015). It determines the safety of women's employment when they are at work or in the office; therefore, workplaces must be secure and women must be provided with the necessary security there. Passengers, security personnel, and other sporadic workers must provide identification documents (driver's license, photo ID, residence confirmation, and fingerprint). Input/output, common hallways, etc., are continuously monitored by CCTV in essential sites or locations for business. However, it is not necessary to jeopardize the workers' security and integrity. Where CCTV is not practical, manned entrances and security deployments are made around-the-clock, according to work schedules, solely at the place. Only authorized employees and individuals can enter the workplace through computerized windows. Superior perimeter barriers to stop human intruders in factories, offices, or institutions. The first or last woman to be lowered in a change at night is either a protection officer or a buddy who rides with the car driver.

WoSApp (Deshpande & Kalita, 2014) WoSApp Women now have a dependable means to contact the police in an emergency thanks to WoSApp. The user can quickly activate the calling feature using the panic button screen. In times of need, the system aids women. Additionally, this program answers the query about the user's location and who to call. Shaking the phone will send the police an emergency message with the user's GPS coordinates and chosen emergency contact.

Abhaya (Chand et al., 2015) this system uses GPS to pinpoint where the troubled person is. The system has modules for user registration, location monitoring, and maintaining a list of emergency contacts. It assists with real-time GPS victim position tracking and calls from the victim's root device are answered by one of the registered contacts. The GPS pinpoints the precise location even while the root device's location changes quickly. When the emergency button is pressed, the user's precise position is sent to the closest police station as well as friends and family members.

Empowering Women (Yarabothu & Thota, 2015) some of the currently available applications provide comprehensive resources for victims of domestic violence as well as a mechanism to get help when needed. Other applications are solely intended for emergency calls when someone may be in danger. However, this application will offer information on domestic violence prevention laws and female health advice. The user must register in order to use the mobile application. The cloud database stores all of the user's data. The user's smart phone's GPS function will pinpoint the victim's precise location. The victim's precise location will be found by the mobile device's GPS function. The victim can transmit messages containing location and time to the police, as well as to family members, with the aid of the emergency call system.

When the phone's power button is pressed twice, the mobile application VithU (Bankar, 2018) sends a message to previously chosen contacts. Every two minutes, the message is sent out with the most recent coordinates and includes the user's GPS location.

When a button on the app screen is clicked, Nirbhaya (n.d) sends a message with the user's GPS location to a list of emergency contacts. Every time the location changes by 300 meters, the coordinates are updated and sent again. Additionally, it is open-source and free, making it simple to modify and develop enabling quick replication of the application in other countries.

With the help of the app Glympse (n.d), users may broadcast their locations with friends and family in real time while employing GPS tracking. There is no registration required for this software, and there are no contacts to keep track of.

Backlash Mahindra faction is the company that made this app. In the past, the consumer had to pay for this app; it wasn't a free service. The "User is in trouble" message is sent by email, GPS, SMS, and GPRS using this software. This app is compatible with mobile devices that can run Java on Android. Additionally, the application will SMS your location and a map.

3. PROPOSED SYSTEM

3.1 Classification of Data

There is a need to access too many sorts of data to produce prediction models, safety suggestions, and empowering tools in order to improve women's safety through AI. The types of data one could require are listed below, along with some potential data sources:

- Crime statistics:
    i. Incident reports: Information on crimes such as harassment, assault, stalking, and other situations involving safety.

  ii. Geographical data: Location information, including the latitude and longitude of incidents.
  iii. Time and date details: Timestamps to investigate event temporal patterns.
- Demographic Data:
  i. Information on the demographics of the people involved in occurrences, including their age and gender.
  ii. Socioeconomic Information: a person's financial situation, degree of schooling, and employment situation.
- Environmental information:
  i. Lighting conditions: Information on the brightness of different regions throughout the day.
  ii. Access to public transit: Closeness to choices for public transportation.
  iii. Public infrastructure: Information on where public amenities like police stations and emergency call boxes are located.
- Social media information:
  i. Social media posts: Public tweets, Facebook posts, or Instagram photos with safety-related keywords.
  ii. Sentiment analysis: Text data analysis to determine how people feel about certain safety problems.
- Information about user interactions with safety programs, such as user location and activities made, is recorded in interaction logs.
  i. User-submitted comments and complaints of safety-related issues.
- Weather information:
  i. Weather conditions: Information on weather patterns that may have an impact on safety (such as visibility at night).
- Information on community engagement:
  i. Community survey responses: Community members' opinions and concerns about safety.
  ii. Participation information: Details on community involvement initiatives and safety-related events.
- Historical Data: Records of incidences and trends in crime in particular places from the past.
- Geospatial Data: Maps, boundaries, and other geospatial details pertaining to areas of interest are included in geographic information system (GIS) data.
- Information on Educational Resources: Self-defence skills, safety advice, and resources for women's safety.

It's crucial to remember that the collection and use of this data must strictly adhere to ethical standards and privacy laws. Data anonymization and security has to be given top priority. Additionally advantageous are partnerships with communities for data collection and engagement initiatives as well as cooperation with pertinent authorities and organizations to gain access to official crime data. Combining various data sources can yield insightful information that can be used to create machine learning models that forecast high-risk regions, provide safety advice, and arm women with knowledge and tools to improve their safety and wellbeing.

3.2 AI-Driven Components

In order to provide comprehensive assistance and empower women, a proposed system for a smart and empowering approach to women's safety utilizing artificial intelligence (AI) would combine several AI-driven components. Here is how such a system might work (Vijaylashmi et al., 2019):

- Applications for Personal Safety- Apps for personal safety have drawn attention because of their potential to improve the safety of women. These apps frequently contain AI capabilities like danger assessment, emergency notifications, and real-time location monitoring. The efficiency and usability of these applications have been studied in research.
- Policy prediction -Studies have looked into using artificial intelligence (AI) and machine learning algorithms to predict and prevent crimes against women, such as sexual assault and domestic abuse. The reliability of predictive models and ethical issues have been the focus of research.
- Safety Hotlines and Chatbots- AI-powered chatbots and hotlines have being investigated as easily accessible and private ways for women to report incidents and get information. User experiences, chatbot efficacy, and the incorporation of AI in helplines have all been studied in research.
- Crowd sourced Reporting and Data- Platforms that gather information from the public on cases of assault and harassment against women have become more popular. There has been research on using AI to analyse this data to find trends, hotspots, and patterns.
- Safety in Public Transportation -Studies have looked at how AI might improve travel safety for women, including AI-driven surveillance and predictive analytics.
- Intelligent Street Lighting-Research has looked at how well-lit, AI-powered smart streets can increase safety by illuminating dark spaces more effectively and discouraging criminal activity.
- Community Participation-The effects of AI-driven community involvement activities, such as social media monitoring and awareness campaigns, on increasing awareness of and responsiveness to women's safety concerns have been investigated.
- Education and Training- Research has examined the influence of AI-driven training programs and instructional tools on women's safety, as well as how well they teach self-defence techniques and other safety-related information.
- Legal Assistance- Studies have looked into how AI could be used to speed up legal procedures for victims of violence, such as applications for restraining orders and putting victims in touch with legal help.
- Data security and privacy- Strong data protection measures are essential, as ethical and privacy issues relating to the use of AI in women's safety initiatives have been studied.
- Crisis Response Systems- Studies have looked at how AI might improve crisis response systems' effectiveness, including the prompt dispatch of emergency services.
- User Feedback and Continuous Improvement- According to research, it is crucial to collect user feedback in order to iterate and enhance AI-powered safety solutions continuously.

There are arguments regarding ethical issues, biases in AI algorithms, and the requirement for all-encompassing, community-driven approaches, even though these studies and efforts offer insightful information about the potential of AI in ensuring the safety of women. As a result of a dedication to empowering women and fostering safer surroundings through cutting-edge AI-driven solutions, research in this area is dynamic and always changing. The long-term effects, scalability, and intersectionality of AI-based women's safety programs may be explored in future research.

3.3 Approaches to Enhance Women Safety

This study examines the various ways that machine learning might improve the safety of women.

- Predictive Analytics: Machine learning algorithms can examine previous data to identify potentially dangerous areas or times, allowing people and authorities to take preventative action.
- Anomaly Detection: Modern algorithms can instantly spot out-of-the-ordinary actions or circumstances, such harassment or hostility, and send out notifications or take appropriate action.
- Voice-Activated Support: Voice-driven virtual assistants with natural language processing (NLP) capabilities can offer covert assistance and link women in need to resources or emergency services.
- Social Media Monitoring: Machine learning algorithms can search social media sites for indications of threats or abuse, enabling prompt support and intervention.
- Safe Transportation: Machine learning may be used by ride-sharing and public transportation platforms to improve the safety of female passengers and guarantee a safe travel.
- Risk Assessment Tools: Machine learning-based risk assessment tools can provide users with knowledge about the security of locations or circumstances, enabling them to make informed decisions.
- Policy and Advocacy: Machine learning-based data-driven insights can guide resource allocation and policy improvements, increasing women's safety on a larger scale.

To preserve user data and prevent algorithmic biases, the application of machine learning in women's safety must prioritize privacy and ethical considerations. The successful implementation of machine learning solutions adapted to regional and cultural requirements can be ensured by collaborative efforts involving technological specialists, communities, and governments (Akojwar & Kshirsagar, 2016).

3.4 Use of Machine Learning Algorithms

Machine learning algorithms can play a crucial role in enhancing women's safety by predicting high-risk areas, providing safety recommendations, and empowering women with resources. The choice of the algorithm depends on the specific task and data available. Here are some machine learning algorithms and their potential applications in women's safety (Glympse, n.d).

1. Random Forest:
   - Application: Predicting high-risk areas for women's safety based on historical incident data and environmental features.
   - Advantages: Robust and can handle mixed data types; provides feature importance rankings.
2. Gradient Boosting (e.g., XGBoost, LightGBM):
   - Application: Identifying patterns and predicting safety incidents, especially in cases with imbalanced data.
   - Advantages: High predictive accuracy and robustness; can handle imbalanced datasets.
3. Support Vector Machine (SVM):
   - Application: Classification tasks like identifying areas with elevated safety risks.
   - Advantages: Effective for binary classification; can handle non-linear relationships with kernel functions.

4. Neural Networks (Deep Learning):
    - Application: Complex tasks, including natural language processing for analysing social media data or image analysis for identifying unsafe locations.
    - Advantages: High capacity to learn complex patterns; suitable for large datasets.
5. K-Means Clustering:
    - Application: Identifying clusters of safety incidents or high-risk areas based on geographic data.
    - Advantages: Helps discover spatial patterns and group similar incidents or locations.
6. Naive Bayes Classifier:
    - Application: Analysing text data for sentiment analysis and identifying areas with negative sentiment or safety concerns from social media.
    - Advantages: Simple and effective for text classification tasks.
7. Reinforcement Learning:
    - Application: Developing smart safety recommendation systems that adapt and improve based on user feedback.
    - Advantages: Suitable for dynamic environments where actions impact future safety recommendations.
8. Time Series Forecasting (e.g., ARIMA):
    - Application: Predicting temporal patterns in safety incidents to allocate resources effectively.
    - Advantages: Handles time-dependent data, useful for resource allocation and planning.
9. Anomaly Detection Algorithms (e.g., Isolation Forest, One-Class SVM):
    - Application: Detecting unusual and potentially unsafe events or locations.
    - Advantages: Effective at identifying outliers or anomalies in data.
10. Fairness and Bias Mitigation Algorithms:
    - Application: Ensuring that machine learning models do not perpetuate biases and are fair to all demographics.
    - Advantages: Helps in addressing ethical concerns and promoting equity in safety systems.

The algorithm that best meet the unique goals of user initiative on women's safety must be carefully chosen. To effectively deploy any machine learning solution for boosting women's safety, critical phases include pre-processing, feature engineering, model evaluation, and continual improvement.

4. RESULTS AND DISCUSSION

4.1 Predictive Model

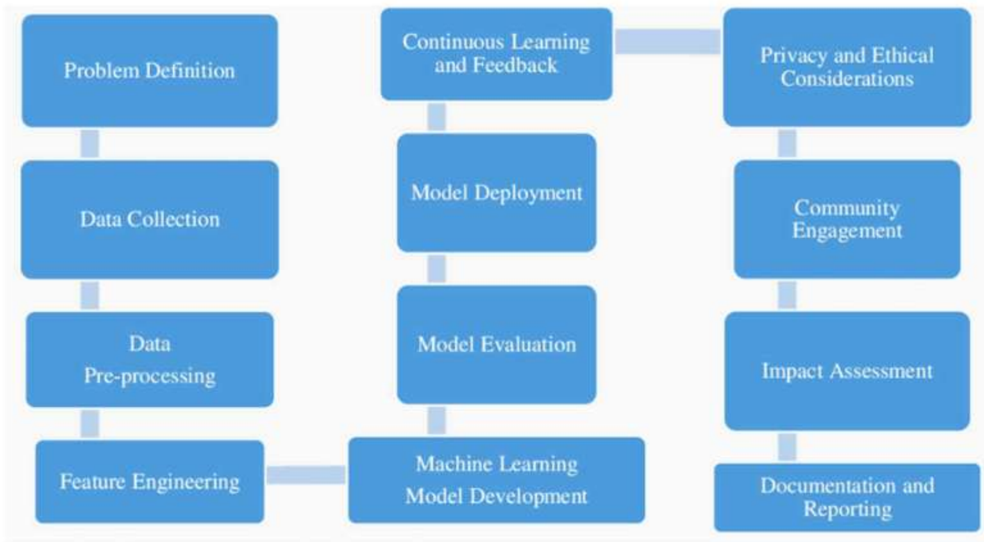
We want to create a prediction model that will enable women to make wise safety decisions, stay away from hazardous situations, and get alerts or safety advice as needed (Vijaylashmi et al., 2019).

1. Data collection: Gather information from a variety of sources, including crime statistics, records of harassment incidents, demographic information, and environmental characteristics (such as well-lit locations and close proximity to public transportation). Make sure the data is spatial, with latitude and longitude details.

2. Data Pre-processing:
    - Deal with duplicates, outliers, and missing values.
    - Normalize numerical characteristics and encode categorical variables.
    - Develop a target variable that will indicate, based on prior instances, whether a location is high-risk or not.
3. Feature Engineering: Engineer relevant features to improve the model's predictive power:
    - Time-based features (day of the week, time of day) to account for temporal patterns.
    - Spatial features (proximity to schools, police stations, etc.).
    - Social media sentiment analysis for the area to capture public sentiment regarding safety.
4. Model Selection: Choose an appropriate machine learning algorithm for classification, considering the nature of the problem and the dataset. Some suitable algorithms include:
    - Random Forest
    - Gradient Boosting (e.g., XGBoost, LightGBM)
    - Support Vector Machine
    - Neural Networks (if the dataset is sufficiently large)
5. Model Training: Split the dataset into training, validation, and test sets. Train the selected model on the training data, optimizing for a relevant evaluation metric (e.g., F1-score or area under the ROC curve).
6. Model Evaluation: Assess the model's performance using metrics such as accuracy, precision, recall, F1-score, and AUC-ROC on the validation and test sets. Fine-tune hyper parameters if needed.
7. Model Deployment: Deploy the trained model as part of a safety application:
    - Integrate the model into a user-friendly interface, such as a mobile app or website.
    - Allow users to input their location or enable GPS tracking for real-time safety recommendations.
    - Provide safety recommendations, warnings, or alerts based on the model's predictions.
8. Continuous Learning and Feedback: Collect user feedback and data on safety incidents reported through the application.
    - Continuously retrain the model with updated data to improve accuracy and relevance.
    - Incorporate user feedback to enhance safety recommendations and features.
9. Privacy and Ethical Considerations:
    - Implement privacy measures to protect user data and ensure data anonymization.
    - Address potential biases in the data and model to ensure fairness and equity.
10. Community Engagement: Engage with the local community to promote the use of the safety application and gather insights for further improvements.
11. Impact Assessment: Regularly assess the impact of the model and safety application on women's safety and empowerment, considering qualitative and quantitative measures.
12. Documentation and Reporting: Maintain documentation of model development, updates, and user feedback. Share findings and success stories with stakeholders and the community.

By creating and implementing such a prediction model, one may improve the safety of women by giving them useful data and tools for making educated decisions and reducing their risk. To maintain the model's efficacy and relevance over time, it is essential to continuously improve it in response to user comments and fresh data. Figure 1 represents the entire structure in terms of Predictive Model.

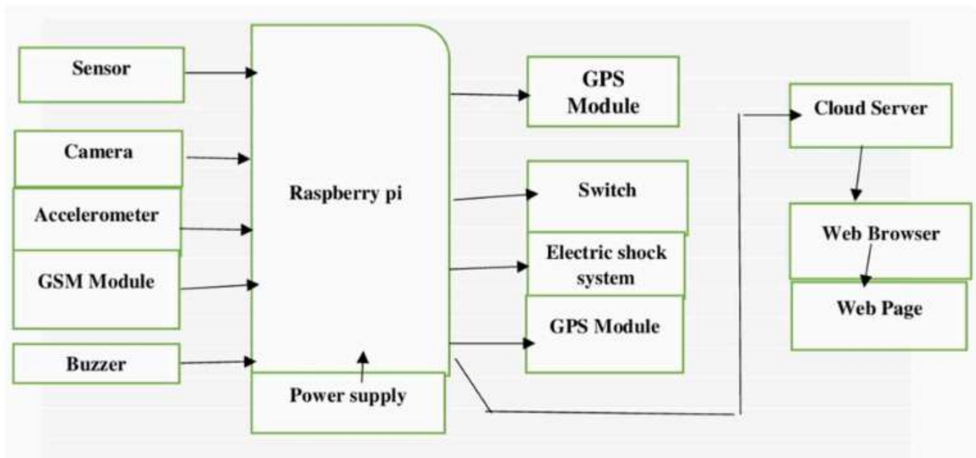
Figure 1. Predictive Model



4.2 Proposed Model Using IoT

Women have historically been the most oppressed and abused group in society, and in recent years, women's welfare has been a major issue. Numerous startling and unexpected statistics can be found. Society is gradually becoming more aware of women's obligations. To create a security framework in this area, some strategic creativity was required. Numerous surveillance tools, including smartphone apps, vibrating jackets, GPS and GSM-based localisation controls, and SOS turn ideation phase. The Battle Back and VithU applications are on the smartphone. They also developed a user-friendly app that, with just two presses of the power button, fulfils a request up to a predetermined length. Every day, this software will update the GPS position. The Back3 software places a phony call, and the video plays starts recording for field documentation (Akojwar & Kshirsagar, 2016; Pravin et al., 2018). Other personal protection technologies that women can deploy in a dangerous scenario include tear gas, flash lights, and shock impacts. These are readily available on the market. While practicing warning code, the specific protection device that employs a portable devices switch is delivered to the given address. It makes use of the most recent generators and detectors for hazard identification .Women continue to have a variety of jobs every day. They frequently provide safer alternatives across cultural, political, and commercial barriers SWMS encourages an incident in order to ensure security report to be made right away. We reasoned that sending a request for a pre-emptive alarm would allow us to be appropriately prepared through safety measures. When they install the app, the woman should go through the enrolment process. She would then be registered. Your home will be visited if you have signed in, and in order to be notified, your personal contacts must be registered .Using a spoken accelerometer and pulse sensory data, this device is a fully integrated, powerful AI system that can specifically identify an accuser's situation. It is run on a Raspberry Pi. It offers an immediate surveillance system, such as a tear gas personality device and an electrical shocker. This automated approach is best suited for any situation, not just the conventional, manually actuated device. This technology offers high levels of stability and dependability, can work as a safe route for women to fly alone, and has the potential to serve as a personal defence aid similar to Voice Search (P & Ramesh Kumar, 2015). Figure 2 displays the artificial intelligence of the lady security system.

Figure 2. Block diagram for proposed model

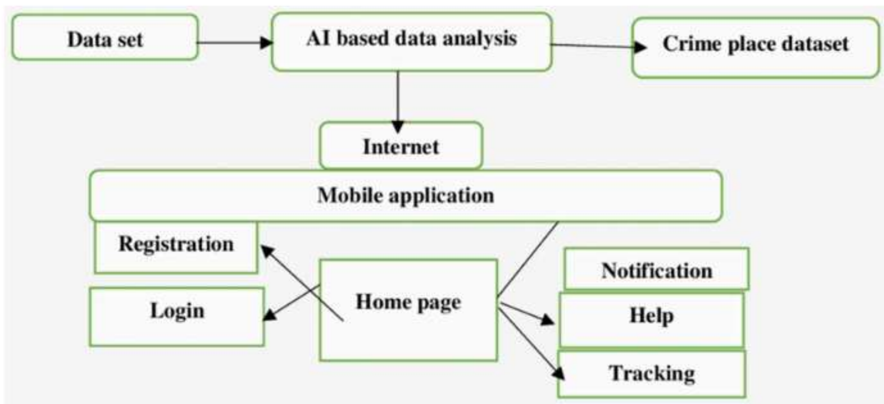


A cross-platform utility for the raspberry pi is loaded with the Raspbian OS and a 16GB memory stick. Use a keyboard, mouse, VNC viewer, display, and SSH to communicate with the Raspberry Pi. LX terminal instructions have been used to connect with them. Among the A USB microphone is coupled with a USB pi port. The GSM module, GPS module, and raspberry pi are all connected to the pi respiration rates and accelerometers monitor they communicate with the raspberry pi's python programs. Jasper is a highly adaptable, Python-based speech software development platform that may also incorporate a wide range of additional functionalities. The majority of text-to-speech and voice-to-text conversions are accessible both offline and online. The conclusions are precise. A wake term will take place. When you receive the wake command, the raspberry pi starts to examine the voice files and convert them to text using the STT engine. The degree of risk is gauged in comparison to terms that have already been established. The pi then acts in accordance with the degree of danger. Late in the evening. Without waiting for the wake, it continually analyses human auditory input. The person communicates with the scented pi while monitoring the accelerometer and pulse detector. After examining the received results, it is compared to the specified estimated quality. If you discover any abnormalities, the voice input will start to search through the wake word and complete the prescribed duty. Each person's pulse intensity is different, so it must be adjusted to fit them all. A person's fitness can also be monitored using their pulse rate. Gas canisters or an electric shock production jacket will occur in the system of time safety. Depending on the severity of the shock risk, jackets are designed to operate wirelessly over WIFI ().

4.3 Application Development

The established spatial data kit checks to see if the current location fits or fails to fit its database if a woman is checking in and going to a new location. If it does, it indicates that the location is in a high-crime area. If she had known where she was most likely to face violence, she would have changed. She would stop visiting there, or under protective measures, she might require training to assist SWMS. She is provided the assistance button on the app if she decides to go to a location that is prone to violence and worries about an outside danger. By pressing the call button, a response will appear. Its latitude is the text that is created. The current situation and the slogan "I am at trouble please help" will be presented. I need your help. The generated notification is forwarded to the saved emergency numbers. In Figure 3, we can see the machine flow.

Figure 3. System flow: Artificial intelligence model



Artificial intelligence is the simulation of general brain functions by machines, primarily personal computers. It is based on the idea that machines ought to think, act, and learn like people. The SVM algorithm includes AI and machine learning. Assignment help Using the Vector Machine Algorithm, categorization is done. The SVM technique in this study aids in the detection of violent zones. After a crime has been detected, the machine can notify the user using this algorithm. In figure 4, represents flow chart when the button is turned, the system is immediately activated milliseconds later. Automatically tracking the accused's whereabouts, alerts for female relations are provided to the sophisticated motion detection device. The crying warning device is activated, producing a siren signal to signal for help. In order to injure the intruder and possibly help the victim escape, electric shock is utilized. With the chosen IP address, streaming live video can assist in managing the victim scenario such that the attacker's face and the surrounding area are easily and quickly recognized (Jude et al., 2021).

4.4 Result Analysis

The proposed system would specifically understand the victim's health and its context by becoming accustomed to user voice, responding to it, and helping user in perilous situations by phoning emergency personnel and providing shock-creation jackets in good time introduce a procedure. If a girl is assaulted by anyone while the jacket button is pressed, four concurrent acts are permitted. Figure 5 emphasizes the message that the girl sends when the emergency touch is activated on a raspberry Pi. Then, because the circle is isolated, an electric shock that only affects the attacker will be generated. To prevent excessive wave propagation inside the electric shock circuit, an opto isolator has been used. The public is promptly alerted to the emergency by pressing the panic button. The camera is activated and will start to broadcast live video with a five second delay or begin to display the current moment with a two second delay. The data is then transmitted to the Raspberry Pi's database and ultimately, via AI, to the cloud server. Figure 5 gives the message sent by Raspberry Pi.

Figure 4. Flow chart

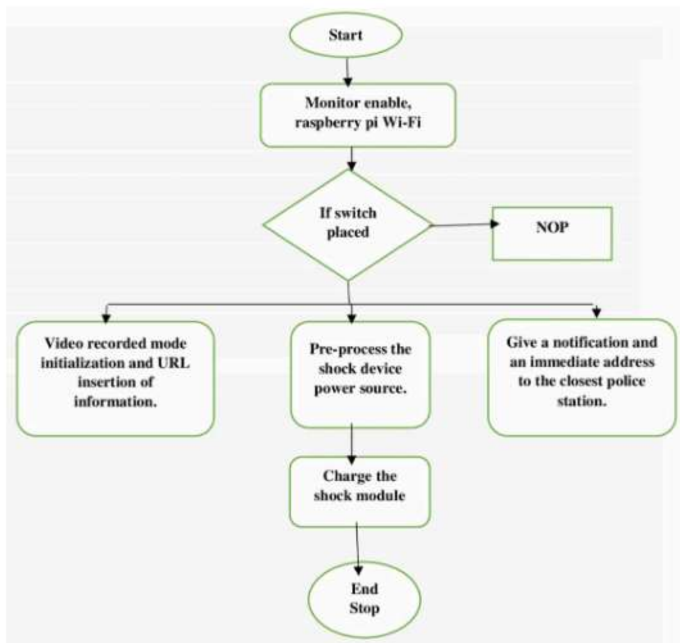
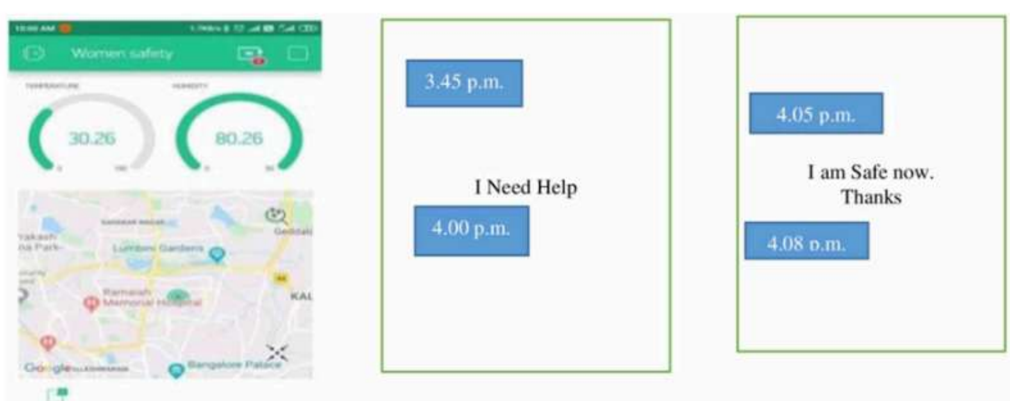


Figure 5. Message sent by Raspberry Pi



5. ADVANTAGES OF THE PROPOSED SYSTEM

The use of artificial intelligence (AI) to implement a smart and empowering approach for women's safety using artificial intelligence (AI) has a number of benefits that can dramatically improve women's safety and wellbeing in a variety of scenarios. Here are a few significant advantages (Jude et al., 2021; Kshirsagar, 2020):

- Enhanced Personal Safety: AI-powered personal safety applications provide women with tools like real-time location tracking and emergency alerts, improving their ability to respond quickly to potential threats or emergencies.
- Preventive Measures: Predictive policing and risk assessment powered by AI can help identify high-risk areas and times, enabling proactive measures to prevent incidents of violence against women.
- Confidential Reporting: AI-driven chatbots and hotlines offer a confidential and non-judgmental platform for women to report incidents, seek information, and access support services without fear of stigma or retaliation.
- Efficient Resource Allocation: Predictive models assist law enforcement in allocating resources more efficiently, ensuring a quicker response to incidents and improving overall safety in communities.
- Community Engagement and Awareness: AI-driven community engagement initiatives and awareness campaigns can raise public awareness about women's safety concerns and foster collective action to address them.
- Education and Empowerment: AI-powered educational tools and training programs empower women by providing them with knowledge and skills to protect themselves and make informed decisions.
- Rapid Crisis Response: AI streamlines crisis response systems, enabling faster dispatch of emergency services and providing critical information to first responders.
- Data-Driven Decision-Making: AI analyses crowd sourced data to identify trends and patterns, helping organizations and authorities make data-driven decisions to improve safety measures.
- Privacy and Security: AI can enhance data privacy and security, ensuring that sensitive information is protected while providing the necessary support and resources to women.
- User-Centric Design: AI-driven solutions can be tailored to meet the specific needs and preferences of users, creating more user-friendly and accessible platforms.
- Continuous Improvement: User feedback mechanisms integrated into AI systems allow for continuous improvement, ensuring that safety solutions remain effective and relevant over time.
- Scalability and Accessibility: AI-based systems can be scaled to reach a broader audience, making safety resources and support accessible to women in diverse geographic and cultural contexts.
- Crisis Prevention: AI can identify potential risks and threats before they escalate, contributing to the prevention of violence and harassment against women.
- Efficiency and Resource Optimization: AI streamlines processes, reducing the time and effort required for tasks such as legal assistance and incident reporting, making these resources more readily available to women in need.
- Reduced Bias and Discrimination: By relying on data and algorithms, AI systems can reduce human biases in decision-making, leading to more equitable and fair responses to incidents.

In summary, a smart and empowering approach for women's safety using AI not only addresses immediate safety concerns but also fosters a culture of empowerment, awareness, and prevention. It leverages technology to provide women with the tools and resources they need to lead safer lives and contributes to building more inclusive and secure communities.

6. SOCIAL WELFARE OF THE PROPOSED SYSTEM

Implementing a smart and empowering approach for women's safety using artificial intelligence (AI) can have significant implications for social welfare. Here are ways in which this approach can contribute to social welfare:

- Reduced Violence and Harassment: By leveraging AI for predictive policing and real-time monitoring, the approach can lead to a reduction in incidents of violence and harassment against women. This, in turn, contributes to the physical and psychological well-being of women, leading to improved social welfare.
- Enhanced Access to Support Services: AI-powered platforms can provide women with easier access to support services, including counselling, legal aid, and medical assistance. This ensures that women receive the necessary care and support, improving their overall welfare.
- Empowerment and Education: AI-driven educational tools and training programs empower women with knowledge and skills to protect themselves. Empowered women are more likely to have higher self-esteem, make informed decisions, and actively participate in society, which positively impacts social welfare.
- Community Engagement: AI can facilitate community engagement and awareness campaigns. When communities are educated about women's safety issues and actively participate in addressing them, social welfare improves as safety becomes a collective responsibility.
- Reduced Fear and Anxiety: As women feel safer in their communities due to AI-enhanced safety measures, they experience reduced fear and anxiety. Improved mental health contributes to better overall welfare.
- Efficient Resource Allocation: AI assists in efficient resource allocation, ensuring that law enforcement and support services are directed to areas and individuals in need. This reduces wastage and optimizes the use of public resources, benefiting society as a whole.
- Data-Driven Decision-Making: AI's ability to analyse data helps authorities make informed decisions about safety measures. Evidence-based policies and interventions are more likely to have a positive impact on social welfare.
- Reduced Gender Disparities: An AI-driven approach can help reduce gender disparities in safety by ensuring that safety measures are tailored to the specific needs and experiences of women. This promotes greater gender equality and overall social welfare.
- Accessibility and Inclusivity: AI systems can be designed to be accessible and inclusive, ensuring that women from diverse backgrounds, including those with disabilities or from marginalized communities, can benefit from these safety measures, further enhancing social welfare.
- Prevention and Long-Term Impact: AI's ability to predict and prevent incidents of violence means that it can contribute to long-term improvements in social welfare by reducing the cycle of violence and its associated costs.

- Privacy and Data Protection: Ensuring that AI systems protect user data and privacy enhances the overall sense of security and trust, contributing to improved social welfare.

In conclusion, a smart and empowering approach for women's safety using AI has the potential to significantly enhance social welfare by reducing violence, empowering women, engaging communities, and promoting a safer, more inclusive society. It contributes to the well-being and quality of life of women and the broader community, ultimately leading to a more equitable and prosperous society ().

7. FUTURE ENHANCEMENT

The future of a smart and empowering approach for women's safety using artificial intelligence (AI) holds tremendous potential for innovation and improvement. Here are some key areas where enhancements and developments are likely to occur:

- Advanced Predictive Models: Enhance predictive policing models with more sophisticated algorithms and data sources. Incorporate real-time social media and geospatial data for more accurate risk assessment.
- IoT Integration: Integrate the Internet of Things (IoT) devices such as wearables and smart home security systems with AI to provide continuous monitoring and immediate alerts in case of emergencies.
- Virtual Reality (VR) and Augmented Reality (AR): Create immersive VR and AR training simulations for women to practice self-defence techniques and situational awareness in a safe environment.
- Multilingual and Multicultural Solutions: Design AI systems that are multilingual and culturally sensitive to cater to the needs of diverse communities and regions worldwide.
- Blockchain for Data Security: Use blockchain technology to enhance the security and privacy of user data, ensuring that sensitive information remains confidential and tamper-proof.

As technology continues to advance, these enhancements will contribute to more sophisticated, inclusive, and effective AI-driven solutions for women's safety, ultimately creating a safer and more empowering environment for women worldwide (Kshirsagar, 2020).

8. CONCLUSION

In conclusion, a smart and empowering approach for women's safety using artificial intelligence (AI) promises a paradigm-shifting route to building safer, more inclusive, and egalitarian communities. Women's safety programs that use AI technology provide a bright future where they may live more confidently knowing that protective safeguards and support systems are in place to protect their wellbeing. This strategy uses real-time tracking, predictive policing, and emergency response systems to increase personal safety. By giving them access to necessary resources, education and training tools, and a platform to speak up and share their stories, it empowers women. Additionally, AI-driven solutions encourage civic participation, awareness, and shared accountability, as well as effective resource management and data-driven decision-making on the part of authorities. In order to maintain the confidentiality of sensitive information, privacy and security are given priority. While the path to a more secure and empowered environment for women utilizing AI is encouraging, it is crucial to keep an eye out for issues of ethics, fairness, and diversity. The success of this strategy depends on collaboration between governments, technological corporations, community organizations, and private citizens. In conclusion, an intelligent and powerful AI-based strategy for women's safety not only answers urgent safety concerns but also demonstrates a dedication to creating a society in which women can flourish, contribute, and participate fully without fear of violence or harassment. It embodies the values of human rights, equality, and empowerment, and it is a goal worth working toward in order to improve everyone's future.

REFERENCES

Akojwar, S. & Kshirsagar, P. (2016). A Novel Probabilistic-PSO Based Learning Algorithm for Optimization of Neural Networks for Benchmark Problems. WSEAS International conference On Neural Network, Rome, Italy.

Bankar, S. A. (2018). Kedar Basatwar, Priti Divekar, Parbani Sinha, Harsh Gupta, "Foot Device for Women Security. 2nd International Conference on Intelligent Computing and Control System. IEEE. doi:10.1109/ICCONS.2018.8662947.10.1109/ICCONS.2018.8662947

Chand, D., Nayak, S., Bhat, K. S., Parikh, S., Singh, Y., & Kamath, A. (2015). A Mobile Application for Women's Safety: WoSApp. IEEE Region Conference. IEEE. doi:10.1109/TENCON.2015.737317110.1109/TENCON.2015.7373171

Choudhary, Y., Upadhyay, S., Jain, R., & Chakrabortey, A. (2017). Women Safety Device. IJARSE 06(5).

Chougula, B., Archananaik, M., Monu, P., & Patil, P. (2014). Smart Girls' Security System. Priyanka Das,IJAIEM, 03(4), 2319–4847.

Deshpande, M., & Kalita, K. (2014). Ramachandran. M. International Journal of Applied Engineering Research: IJAER, 9(23), 21975–21992.

Glympse. (n.d.). Real-Time Geo-Location Technology. Glympse Corp..

Jude, A. B., Singh, D., & Islam, S. (2021). An Artificial Intelligence Based Predictive Approach for Smart Waste Management. Wireless Personal Communications. doi:10.1007/s11277-021-08803-7 doi:10.1007/s11277-021-08803-7

Kaur, S., Sharma, S., Jain, U., & Raj, A. (2016). Voice command system uding paspberry pi. Advanced Computational Intelligence: International Journal, 3(3).

Kshirsagar, P. (2020). Brain Tumor Classification and Detection Using Neural Network. Proceedings of the 2013 Fourth International Conference on Computing, Communications and Networking Technologies (ICCCNT), (pp. 83–88). IEEE.

Nirbhaya. (n.d.). Home. Nirbhaya.

Pradeep, P., Edwin Raja Dhas, J., & Ramachandran, M. (2015). International Journal of Applied Engineering Research: IJAER, 10(11), 10392–10396.

Pravin K, Akojwar S. (2016). IEEE SCOPES International Conference. IEEE.

Pravin, K., & Akojwar, S. (2016d). Prediction of neurological disorders using PSO with GRNN. IEEE SCOPES International Conference, Paralakhemundi, Odisha, India.

Pravin, R. K., Akojwar, S. G., & Bajaj, N. D. (2018). A hybridized neural network and optimization Algorithms for prediction and classification of neurological disorders. International Journal of Biomedical Engineering and Technology, 28(4), 307–321. doi:10.1504/IJBET.2018.095981 doi:10.1504/IJBET.2018.095981

Ramesh Kumar, P. (2015). Location Identification of the Individual based on Image Metadata. Procedia Computer Science 8, 451 – 454 (2016).

Sangoi, V. B. (2014). International Journal of Current Engineering and Technology, 4(5).

Vijaylashmi, B., Renuka, S., Chennur, P., & Sharangowda, P. (2019). Female Safety Gadget using GPS & GSM Module. JRET International Journal of Research in Engineering and Technology, 6(5), 2319-1163.

Yarabothu, R., & Thota, B. (2015). Abhaya: An Android App For The Safety Of Women. IEEE 12th India International Conference, Electronics, Energy, Environment, Communication, Computer, Control, At Jamia Millia Islamia, New Delhi, India. doi:10.1109/INDICON.2015.744365210.1109/INDICON.2015.7443652

Zhou, C. (2017). HongweiJia, Zhicai Juan, Xuemei Fu, and Guangnian Xiao. IEEE Transactions on Intelligent Transportation Systems, 18(8).